

Lecture 16

Gyroscopic Motions

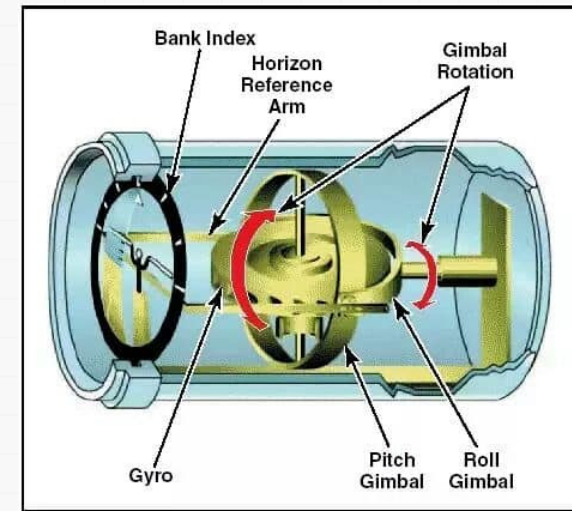
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MC2103
Fall 2026



16 Gyroscopic Motion

Introduction

So far, we have studied kinematics and kinetics for rigid bodies in 3D space. Therefore, we can describe complicated motion of rigid bodies in 3D space. In this lecture, we will examine the motion of gyroscope.

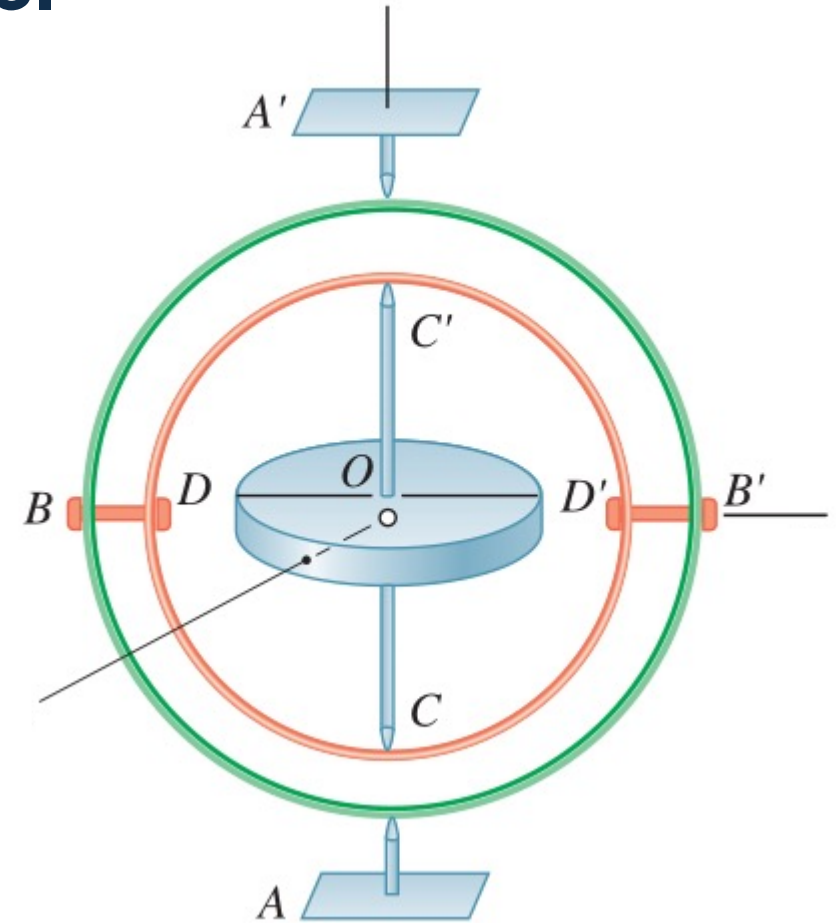


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Model of a Gyroscope

⚙️ Consider the simplified gyroscope model

- Define reference frames
- Assign coordinate systems
- Assign generalized coordinates
- Define direction cosine matrix
- Compute the angular velocity of the rotor
- Compute the angular momentum
- Derive the equations of motion

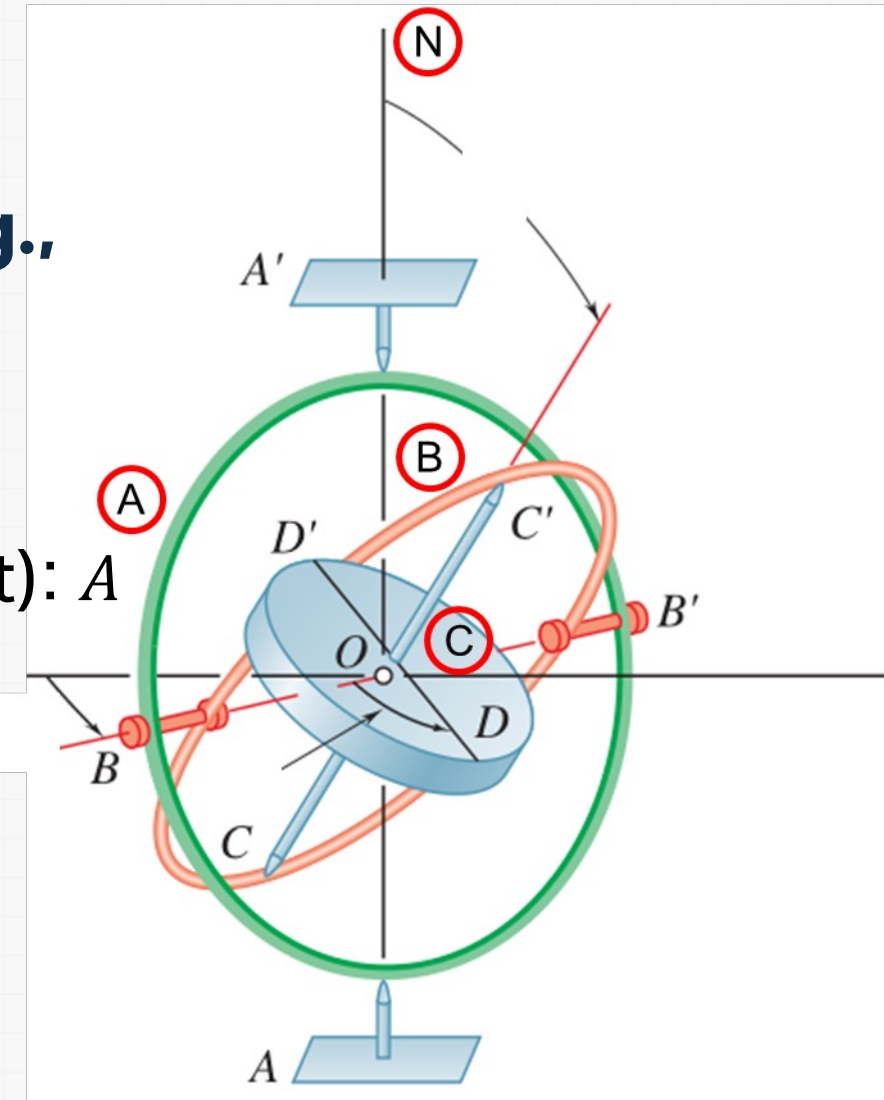


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Reference Frames

⚙️ Identify moving parts for each DOF (e.g., simple rotation)

- Define Newtonian reference frame: N
- Define 1st moving reference frame (outermost): A
- Define 2nd moving reference frame: B
- Define 3rd moving reference frame: C



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Coordinate Systems

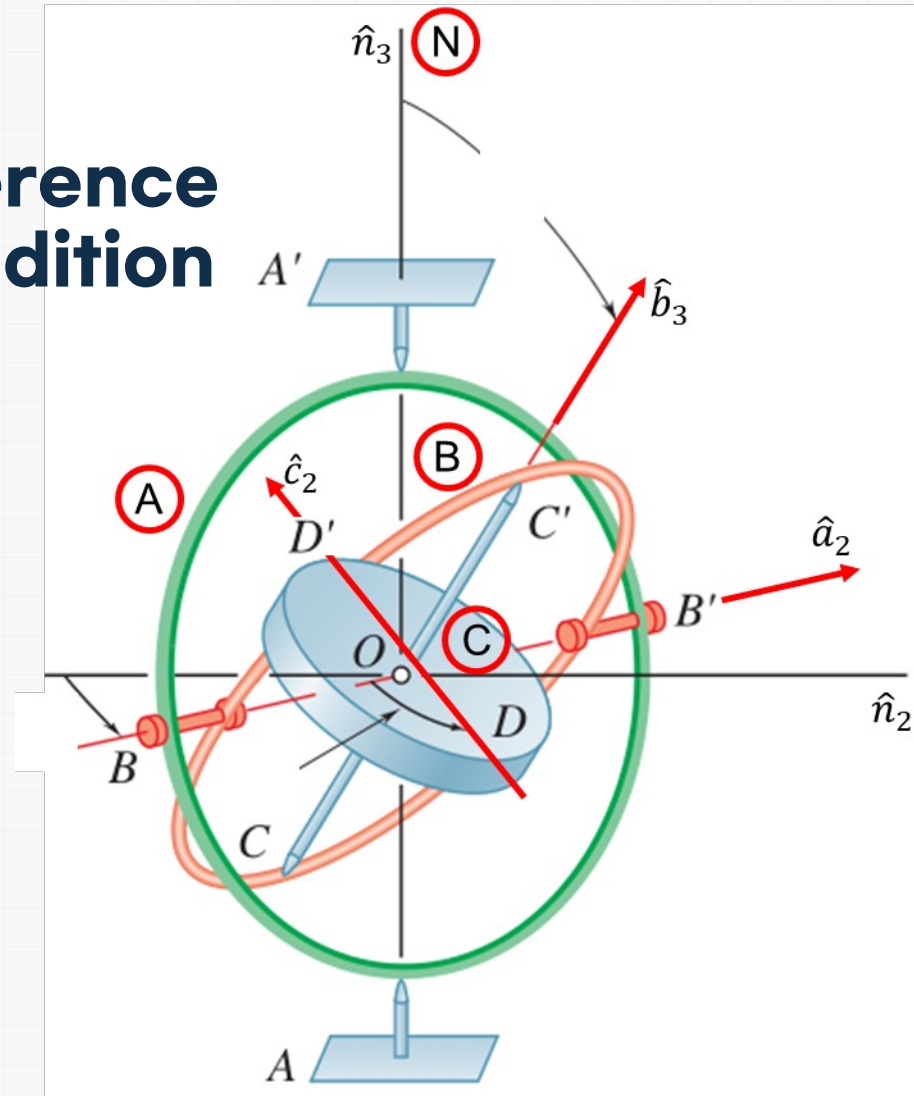
⚙️ Assign coordinate systems for all reference frames while satisfying alignment condition

• For N : $O\hat{n}_1\hat{n}_2\hat{n}_3$

• For A : $O\hat{a}_1\hat{a}_2\hat{a}_3$

• For B : $O\hat{b}_1\hat{b}_2\hat{b}_3$

• For C : $O\hat{c}_1\hat{c}_2\hat{c}_3$

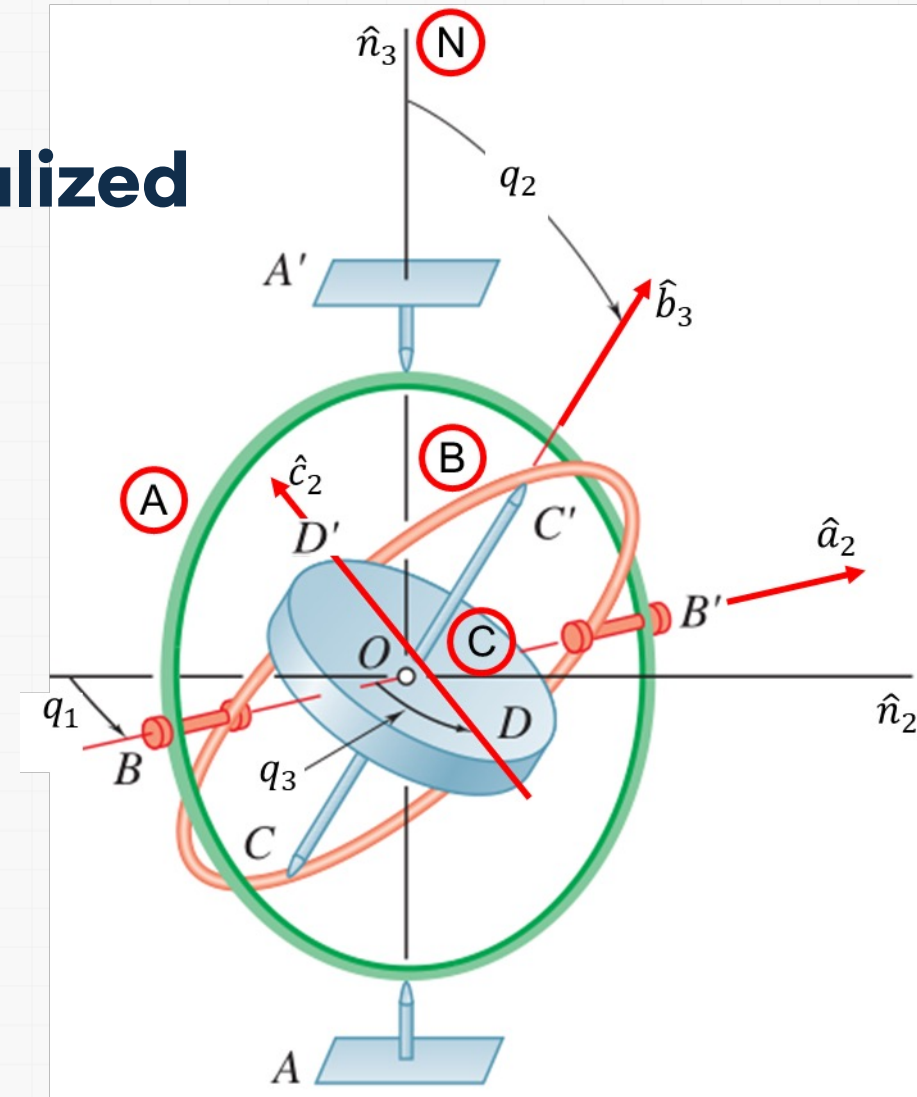


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Generalized Coordinates

⚙️ Define the (minimally required) generalized coordinates

- For $0\hat{n}_1\hat{n}_2\hat{n}_3 \rightarrow 0\hat{a}_1\hat{a}_2\hat{a}_3$: q_1
- For $0\hat{a}_1\hat{a}_2\hat{a}_3 \rightarrow 0\hat{b}_1\hat{b}_2\hat{b}_3$: q_2
- For $0\hat{b}_1\hat{b}_2\hat{b}_3 \rightarrow 0\hat{c}_1\hat{c}_2\hat{c}_3$: q_3



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Direction Cosine Matrix

⚙️ Generate direction cosine tables (using rotation matrices)

	$\hat{a}_1$	$\hat{a}_2$	$\hat{a}_3$
$\hat{n}_1$	$\cos q_1$	$-\sin q_1$	0
$\hat{n}_2$	$\sin q_1$	$\cos q_1$	0
$\hat{n}_3$	0	0	1

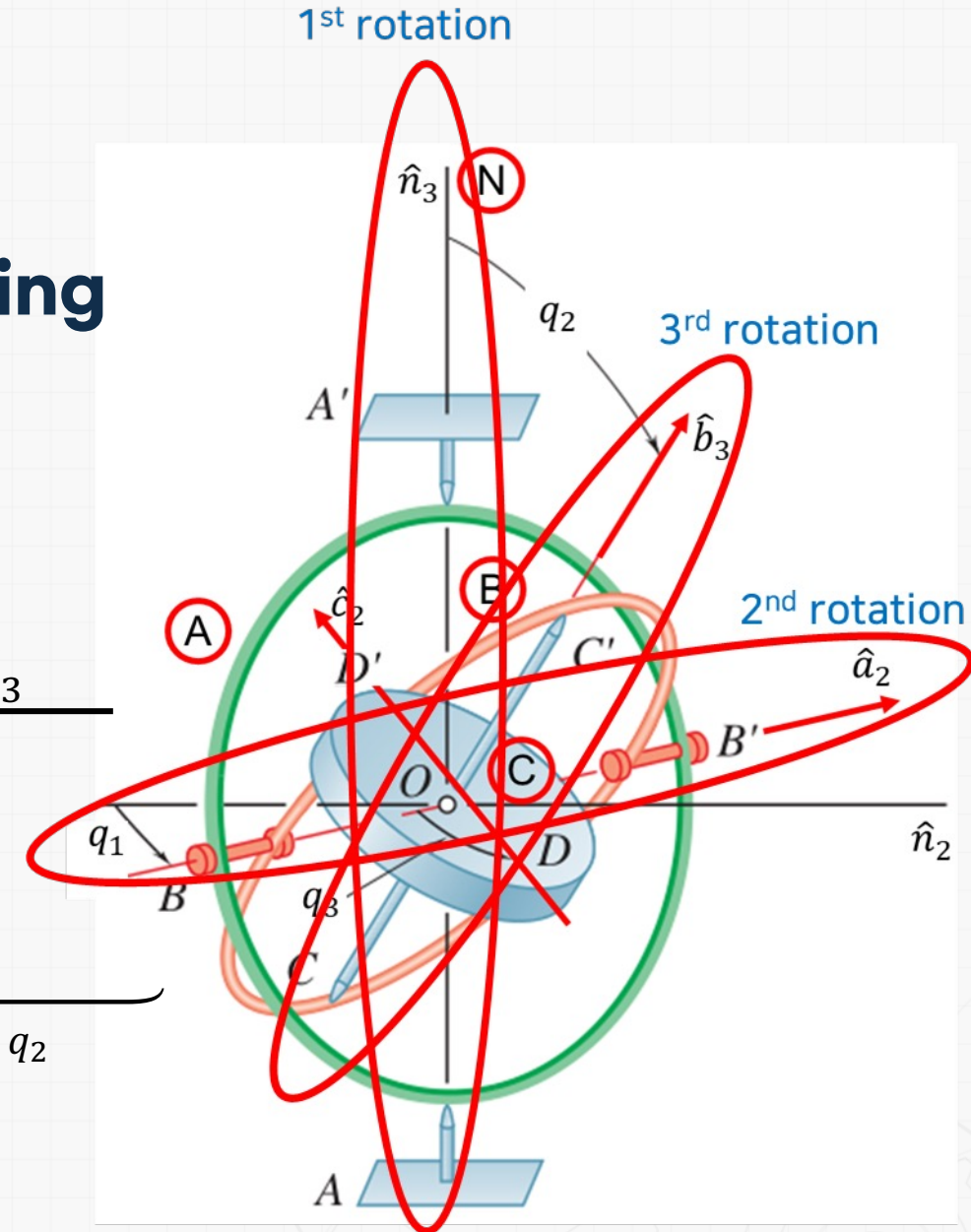
Rotation about z axis by q_1

	$\hat{c}_1$	$\hat{c}_2$	$\hat{c}_3$
$\hat{b}_1$	$\cos q_3$	$-\sin q_3$	0
$\hat{b}_2$	$\sin q_3$	$\cos q_3$	0
$\hat{b}_3$	0	0	1

Rotation about z axis by q_3

	$\hat{b}_1$	$\hat{b}_2$	$\hat{b}_3$
$\hat{a}_1$	$\cos q_2$	0	$\sin q_2$
$\hat{a}_2$	0	1	0
$\hat{a}_3$	$-\sin q_2$	0	$\cos q_2$

Rotation about z axis by q_2



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Angular Velocity of Rotor

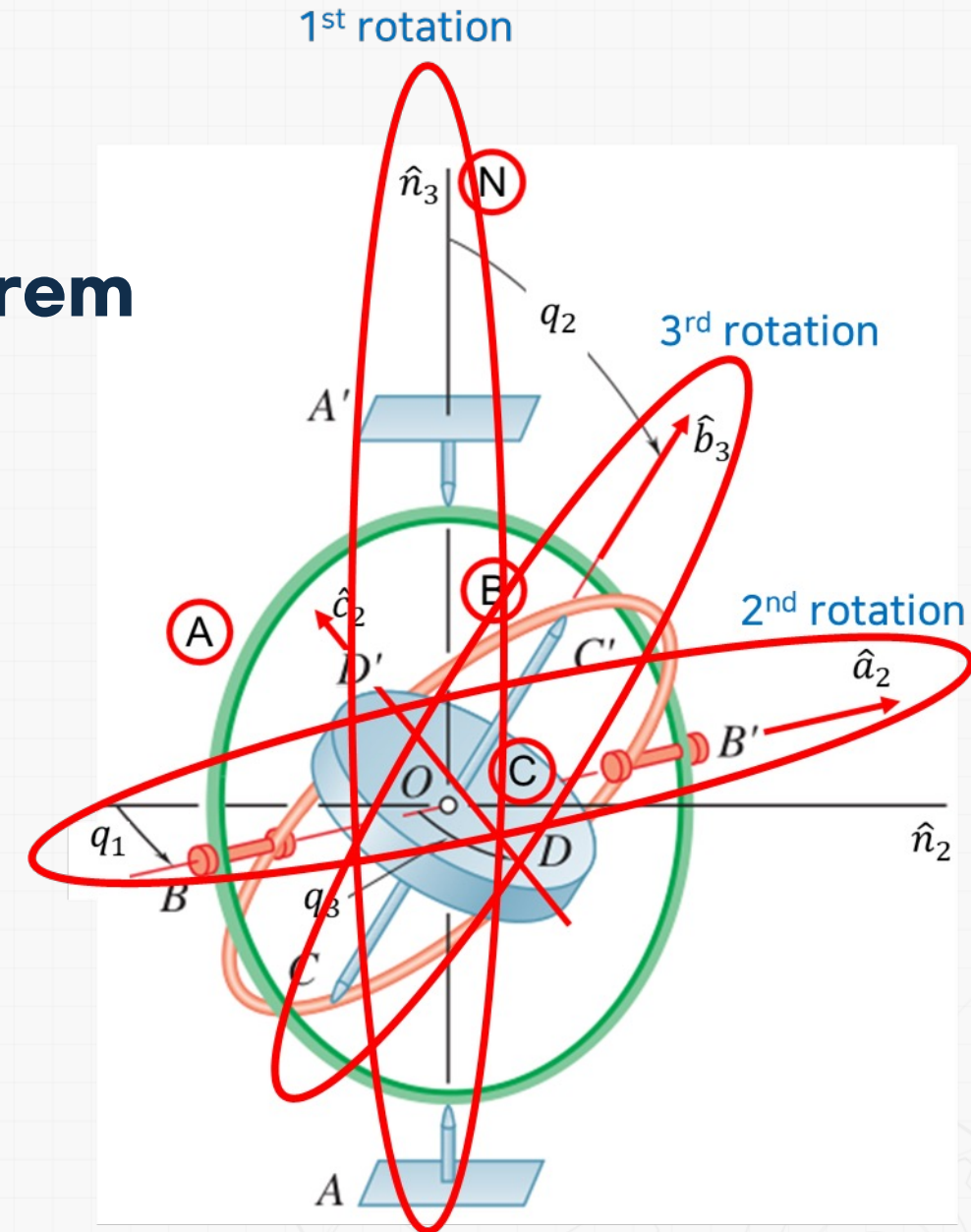
⚙️ Apply angular velocity addition theorem

$${}^N\vec{\omega}^A = \dot{q}_1 \hat{n}_3$$

$${}^A\vec{\omega}^B = \dot{q}_2 \hat{a}_2$$

$${}^B\vec{\omega}^C = \dot{q}_3 \hat{b}_3$$

$$\begin{aligned} {}^N\vec{\omega}^C &= {}^N\vec{\omega}^A + {}^A\vec{\omega}^B + {}^B\vec{\omega}^C \\ &= \dot{q}_1 \hat{n}_3 + \dot{q}_2 \hat{a}_2 + \dot{q}_3 \hat{b}_3 \end{aligned}$$



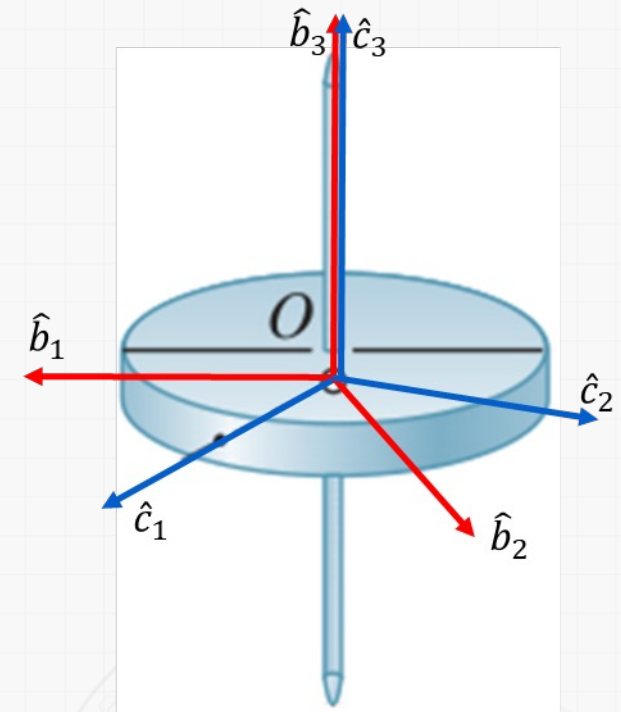
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Momentum of Inertia of Rotor (I)

⚙️ Use body reference frame of the rotor

- Moment of inertia of rotor about COM

$$I_G = \begin{bmatrix} \frac{1}{4}mr^2 & 0 & 0 \\ 0 & \frac{1}{4}mr^2 & 0 \\ 0 & 0 & \frac{1}{2}mr^2 \end{bmatrix}$$



$$I_x = I_y = \frac{1}{4}mr^2$$
$$I_z = \frac{1}{2}mr^2$$

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Angular Velocity of Rotor (I)

⚙️ Use body reference frame of the rotor

• Angular velocity

$$\begin{aligned}
{}^N\vec{\omega}^C &= \dot{q}_1 \hat{n}_3 + \dot{q}_2 \hat{a}_2 + \dot{q}_3 \hat{b}_3 \\
&= \dot{q}_1 \hat{a}_3 + \dot{q}_2 \hat{b}_2 + \dot{q}_3 \hat{c}_3 \\
&= \dot{q}_1 (-\sin q_2 \hat{b}_1 + \cos q_2 \hat{b}_3) \\
&\quad + \dot{q}_2 (\sin q_3 \hat{c}_1 + \cos q_3 \hat{c}_2) + \dot{q}_3 \hat{c}_3 \\
&= \dots \\
&= (-\dot{q}_1 \sin q_2 \cos q_3 + \dot{q}_2 \sin q_3) \hat{c}_1 \\
&\quad + (\dot{q}_1 \sin q_2 \sin q_3 + \dot{q}_2 \cos q_3) \hat{c}_2 \\
&\quad + (\dot{q}_1 \cos q_2 + \dot{q}_3) \hat{c}_3
\end{aligned}$$

	$\hat{a}_1$	$\hat{a}_2$	$\hat{a}_3$
$\hat{n}_1$	$\cos q_1$	$-\sin q_1$	0
$\hat{n}_2$	$\sin q_1$	$\cos q_1$	0
$\hat{n}_3$	0	0	1

Rotation about z axis by q_1

	$\hat{b}_1$	$\hat{b}_2$	$\hat{b}_3$
$\hat{a}_1$	$\cos q_2$	0	$\sin q_2$
$\hat{a}_2$	0	1	0
$\hat{a}_3$	$-\sin q_2$	0	$\cos q_2$

Rotation about z axis by q_2

	$\hat{c}_1$	$\hat{c}_2$	$\hat{c}_3$
$\hat{b}_1$	$\cos q_3$	$-\sin q_3$	0
$\hat{b}_2$	$\sin q_3$	$\cos q_3$	0
$\hat{b}_3$	0	0	1

Rotation about z axis by q_3

16 Gyroscopic Motion

Angular Momentum of Rotor (I)

⚙️ Use body reference frame of the rotor

• Angular momentum

$$\begin{aligned}
{}^N\vec{H}_G^C &= I_G {}^N\vec{\omega}^C \\
&= \begin{bmatrix} \frac{1}{4}mr^2 & 0 & 0 \\ 0 & \frac{1}{4}mr^2 & 0 \\ 0 & 0 & \frac{1}{2}mr^2 \end{bmatrix} \begin{bmatrix} -\dot{q}_1 \sin q_2 \cos q_3 + \dot{q}_2 \sin q_3 \\ \dot{q}_1 \sin q_2 \sin q_3 + \dot{q}_2 \cos q_3 \\ \dot{q}_1 \cos q_2 + \dot{q}_3 \end{bmatrix} \\
&= \frac{1}{4}mr^2(-\dot{q}_1 \sin q_2 \cos q_3 + \dot{q}_2 \sin q_3)\hat{c}_1 \\
&\quad + \frac{1}{4}mr^2(\dot{q}_1 \sin q_2 \sin q_3 + \dot{q}_2 \cos q_3)\hat{c}_2 \\
&\quad + \frac{1}{2}mr^2(\dot{q}_1 \cos q_2 + \dot{q}_3)\hat{c}_3
\end{aligned}$$

Note the abuse of notation

16 Gyroscopic Motion

Equations of Motion (I)

⚙️ Use body reference frame of the rotor

• Moments

$$\begin{aligned}
\vec{M}_G &= \frac{{}^N d}{{}^N dt} {}^N \vec{H}_G^C = \frac{{}^C d}{{}^C dt} {}^N \vec{H}_G^C + {}^N \vec{\omega}^C \times {}^N \vec{H}_G^C \\
&= \frac{1}{4} m r^2 (\dot{q}_1^2 \cos q_2 \sin q_2 \sin q_3 + 2 \dot{q}_1 \dot{q}_3 \sin q_2 \sin q_3 + 2 \dot{q}_2 \dot{q}_3 \cos q_3 - \ddot{q}_1 \sin q_2 \cos q_3 + \ddot{q}_2 \sin q_3) \hat{c}_1 \\
&\quad + \frac{1}{4} m r^2 (\dot{q}_1^2 \cos q_2 \sin q_2 \cos q_3 + 2 \dot{q}_1 \dot{q}_3 \sin q_2 \cos q_3 - 2 \dot{q}_2 \dot{q}_3 \sin q_3 + \ddot{q}_1 \sin q_2 \sin q_3 + \ddot{q}_2 \cos q_3) \hat{c}_2 \\
&\quad + \frac{1}{2} m r^2 (-\dot{q}_1 \dot{q}_2 \sin q_2 + \ddot{q}_1 \cos q_2 + \ddot{q}_3) \hat{c}_3 \\
&= M_1^C \hat{c}_1 + M_2^C \hat{c}_2 + M_3^C \hat{c}_3
\end{aligned}$$

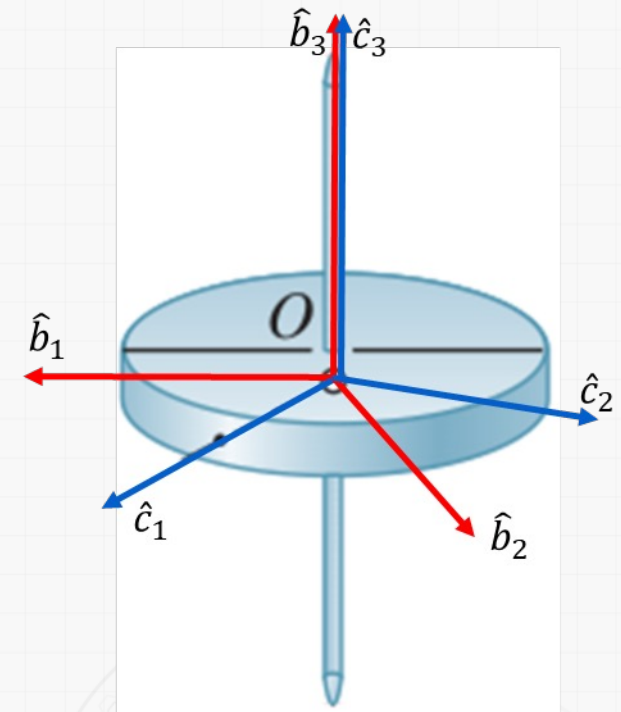
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Momentum of Inertia of Rotor (II)

⚙ Use reference frame B

- Moment of inertia of rotor about COM in $\hat{b}_1\hat{b}_2\hat{b}_3$
= Moment of inertia of rotor about COM in $\hat{c}_1\hat{c}_2\hat{c}_3$
(this is not always true!)

$$I_G = \begin{bmatrix} \frac{1}{4}mr^2 & 0 & 0 \\ 0 & \frac{1}{4}mr^2 & 0 \\ 0 & 0 & \frac{1}{2}mr^2 \end{bmatrix}$$



$$I_x = I_y = \frac{1}{4}mr^2$$
$$I_z = \frac{1}{2}mr^2$$

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Angular Velocity of Rotor (II)

Use reference frame B

Angular velocity

$$\begin{aligned}
 {}^N\vec{\omega}^C &= \dot{q}_1 \hat{n}_3 + \dot{q}_2 \hat{a}_2 + \dot{q}_3 \hat{b}_3 \\
 &= \dot{q}_1 \hat{a}_3 + \dot{q}_2 \hat{b}_2 + \dot{q}_3 \hat{b}_3 \\
 &= \dot{q}_1 (-\sin q_2 \hat{b}_1 + \cos q_2 \hat{b}_3) \\
 &\quad + \dot{q}_2 \hat{b}_2 + \dot{q}_3 \hat{b}_3 \\
 &= -\dot{q}_1 \sin q_2 \hat{b}_1 + \dot{q}_2 \hat{b}_2 + (\dot{q}_1 \cos q_2 + \dot{q}_3) \hat{b}_3
 \end{aligned}$$

	$\hat{a}_1$	$\hat{a}_2$	$\hat{a}_3$
$\hat{n}_1$	$\cos q_1$	$-\sin q_1$	0
$\hat{n}_2$	$\sin q_1$	$\cos q_1$	0
$\hat{n}_3$	0	0	1

Rotation about z axis by q_1

	$\hat{b}_1$	$\hat{b}_2$	$\hat{b}_3$
$\hat{a}_1$	$\cos q_2$	0	$\sin q_2$
$\hat{a}_2$	0	1	0
$\hat{a}_3$	$-\sin q_2$	0	$\cos q_2$

Rotation about z axis by q_2

	$\hat{c}_1$	$\hat{c}_2$	$\hat{c}_3$
$\hat{b}_1$	$\cos q_3$	$-\sin q_3$	0
$\hat{b}_2$	$\sin q_3$	$\cos q_3$	0
$\hat{b}_3$	0	0	1

Rotation about z axis by q_3

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Angular Momentum of Rotor (II)

⚙ Use reference frame B

• Angular momentum

$$\begin{aligned}
 {}^N\vec{H}_G^C &= I_G {}^N\vec{\omega}^C \\
 &= \begin{bmatrix} \frac{1}{4}mr^2 & 0 & 0 \\ 0 & \frac{1}{4}mr^2 & 0 \\ 0 & 0 & \frac{1}{2}mr^2 \end{bmatrix} \begin{bmatrix} -\dot{q}_1 \sin q_2 \\ \dot{q}_2 \\ \dot{q}_1 \cos q_2 + \dot{q}_3 \end{bmatrix} \\
 &= -\frac{1}{4}mr^2 \dot{q}_1 \sin q_2 \hat{c}_1 \\
 &\quad + \frac{1}{4}mr^2 \dot{q}_2 \hat{c}_2 \\
 &\quad + \frac{1}{2}mr^2 (\dot{q}_1 \cos q_2 + \dot{q}_3) \hat{c}_3
 \end{aligned}$$

Note the abuse of notation

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Equations of Motion (II)

⚙ Use reference frame B

• Moments

$$\begin{aligned}\vec{M}_G &= \frac{{}^N d}{{}^N dt} {}^N \vec{H}_G^C = \frac{{}^B d}{{}^B dt} {}^N \vec{H}_G^C + {}^N \vec{\omega}^B \times {}^N \vec{H}_G^C \\ &= \frac{1}{4} m r^2 (2 \dot{q}_2 \dot{q}_3 - \ddot{q}_1 \sin q_2) \hat{b}_1 \\ &\quad + \frac{1}{4} m r^2 (\dot{q}_1^2 \sin q_2 \cos q_2 + 2 \dot{q}_1 \dot{q}_3 \sin q_2 + \ddot{q}_2) \hat{b}_2 \\ &\quad + \frac{1}{2} m r^2 (-\dot{q}_1 \dot{q}_2 \sin q_2 + \ddot{q}_1 \cos q_2 + \ddot{q}_3) \hat{b}_3 \\ &= M_1^B \hat{b}_1 + M_2^B \hat{b}_2 + M_3^B \hat{b}_3\end{aligned}$$

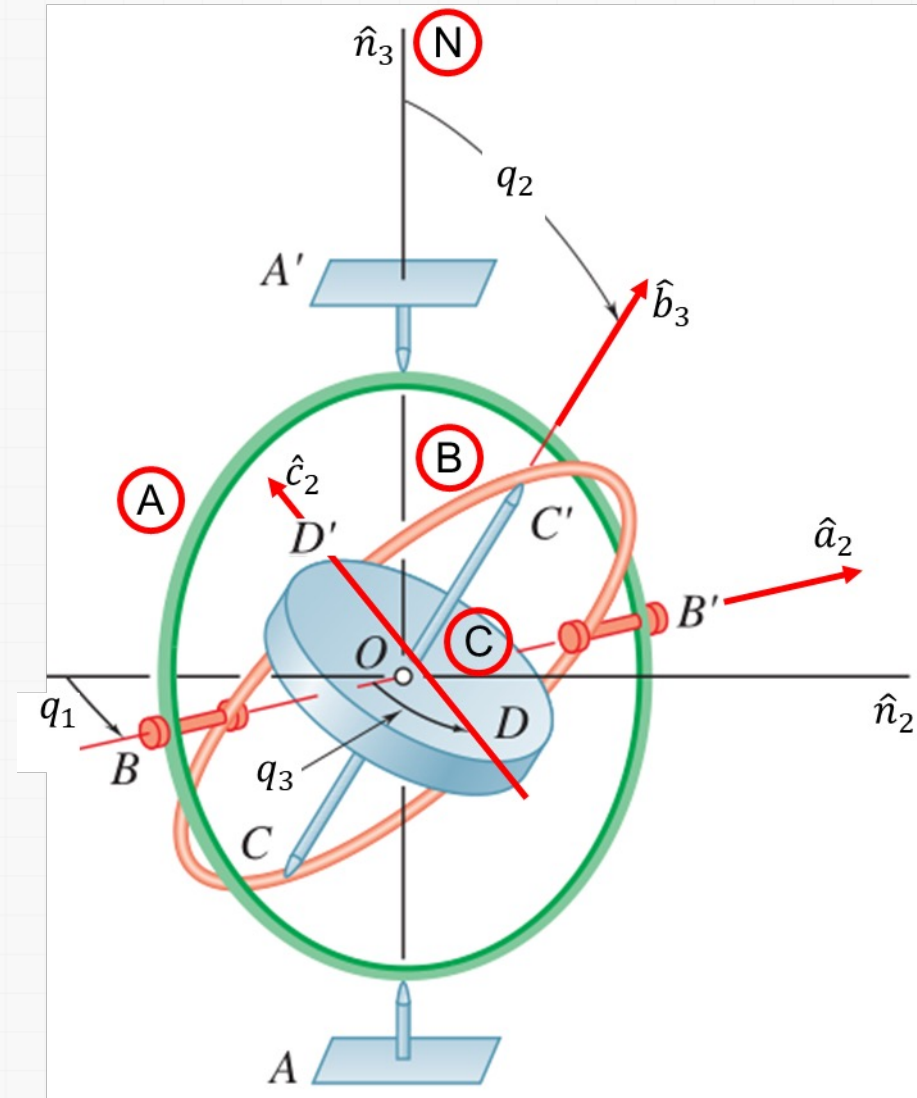
Steady Precession of a Gyroscope

What are the required forces to produce the steady precession?

- We need the following condition.

$$\dot{q}_1 = \text{const}, \dot{q}_2 = 0, \dot{q}_3 = \text{const}$$

$$\ddot{q}_1 = 0, \dot{q}_2 = 0, \ddot{q}_3 = 0$$



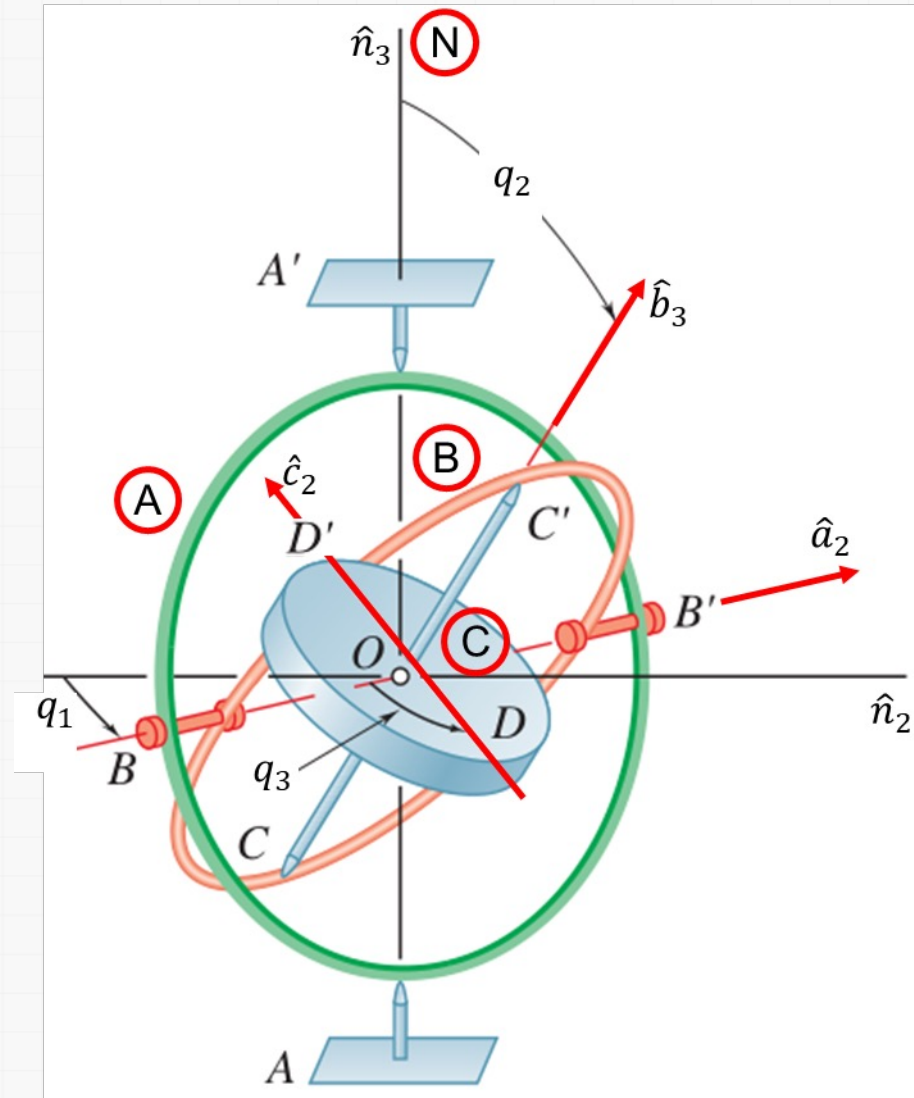
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Steady Precession of a Gyroscope

What are the required forces to produce the steady precession?

- We need the following moments.

$$\begin{aligned}
 \vec{M}_G &= \frac{1}{4}mr^2(2\dot{q}_2\dot{q}_3 - \ddot{q}_1 \sin q_2)\hat{b}_1 \\
 &+ \frac{1}{4}mr^2(\dot{q}_1^2 \sin q_2 \cos q_2 + 2\dot{q}_1\dot{q}_3 \sin q_2 + \ddot{q}_2)\hat{b}_2 \\
 &+ \frac{1}{2}mr^2(-\dot{q}_1\dot{q}_2 \sin q_2 + \ddot{q}_1 \cos q_2 + \ddot{q}_3)\hat{b}_3 \\
 &= \frac{1}{4}mr^2(\dot{q}_1^2 \sin q_2 \cos q_2 + 2\dot{q}_1\dot{q}_3 \sin q_2)\hat{b}_2 \\
 &= M_1^B \hat{b}_1 + M_2^B \hat{b}_2 + M_3^B \hat{b}_3
 \end{aligned}$$



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Steady Precession of a Gyroscope

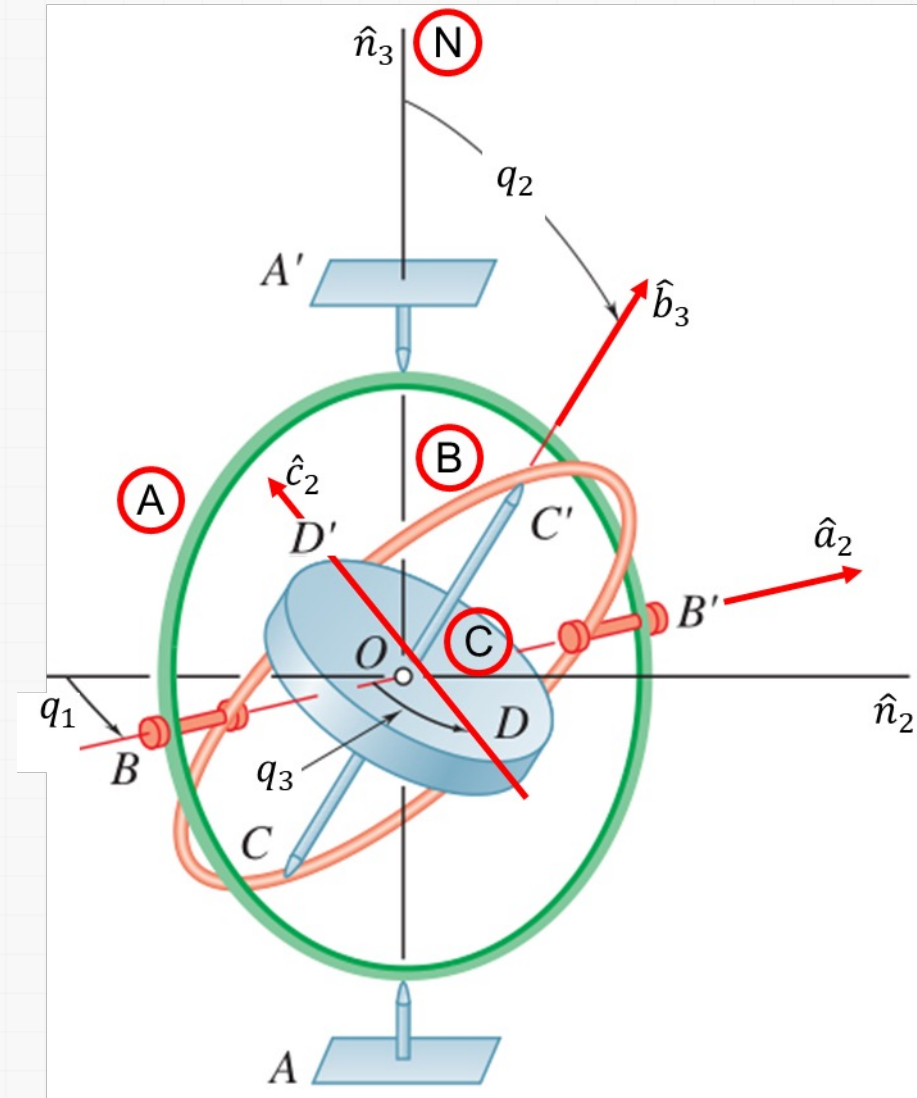
What are the required forces to produce the steady precession?

- We need the following moments.

$$\begin{aligned}\vec{M}_G &= \frac{1}{4}mr^2(\dot{q}_1^2 \sin q_2 \cos q_2 + 2\dot{q}_1\dot{q}_3 \sin q_2)\hat{b}_2 \\ &= M_1^B \hat{b}_1 + M_2^B \hat{b}_2 + M_3^B \hat{b}_3\end{aligned}$$

Therefore, we have the following.

$$\begin{aligned}M_1^B &= M_3^B = 0 \\ M_2^B &= \frac{1}{4}mr^2(\dot{q}_1^2 \sin q_2 \cos q_2 + 2\dot{q}_1\dot{q}_3 \sin q_2)\end{aligned}$$



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Steady Precession of a Gyroscope

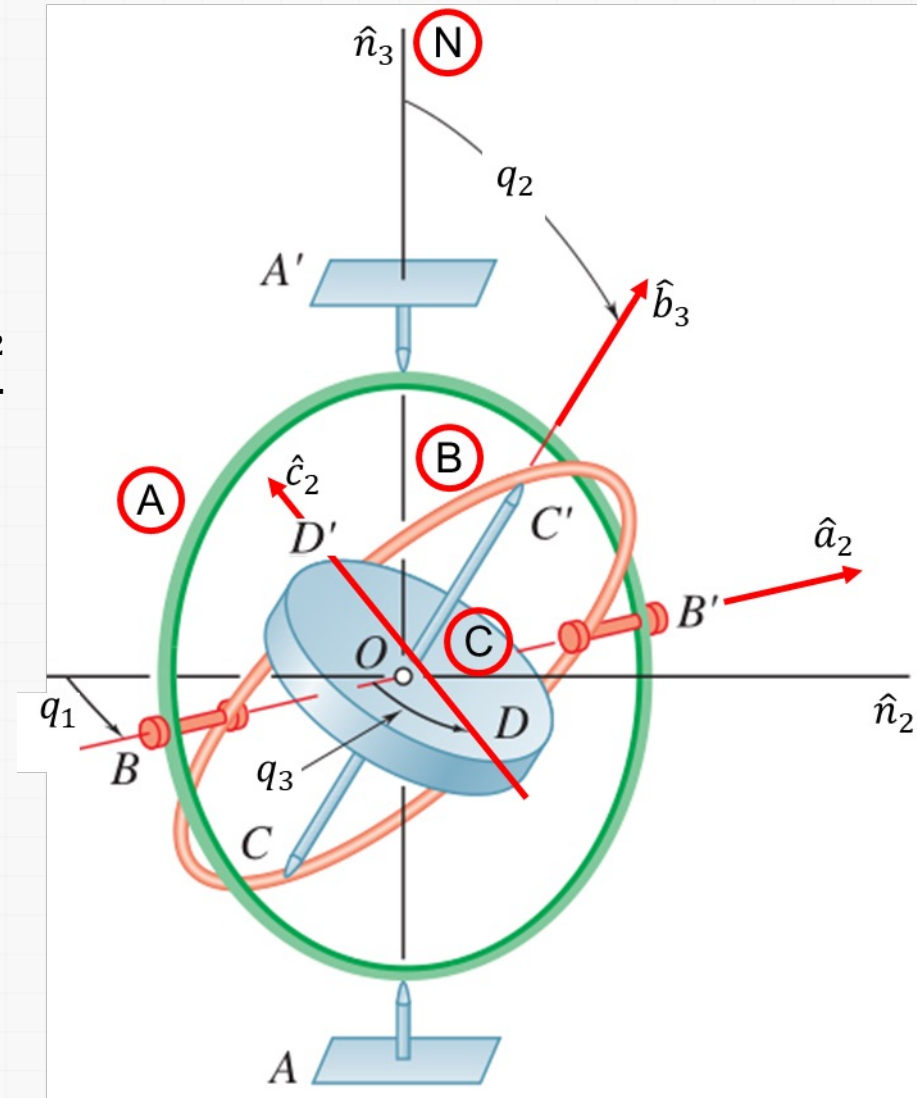
What are the required forces to produce the steady precession?

- If we want a specific precession rate $\dot{q}_1$ (rotation about $\hat{n}_3$) with given spin rate $\dot{q}_3$ and nutation angle q_2 , moment M_2^B should be applied in $\hat{b}_2$ direction which is perpendicular to both spin axis and precession axis.

$$M_2^B = \frac{1}{4} m r^2 (\dot{q}_1^2 \sin q_2 \cos q_2 + 2 \dot{q}_1 \dot{q}_3 \sin q_2)$$

How can we apply $M_2^B (\neq 0)$ while satisfying $M_1^B = M_3^B = 0$?

Put a small weight at D on a reference frame B



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Usage of a Gyroscope

Gyroscopes are used as stabilizers.

- If the precession axis and the spin axis are perpendicular, that is, $q_2 = 90^\circ$, we have

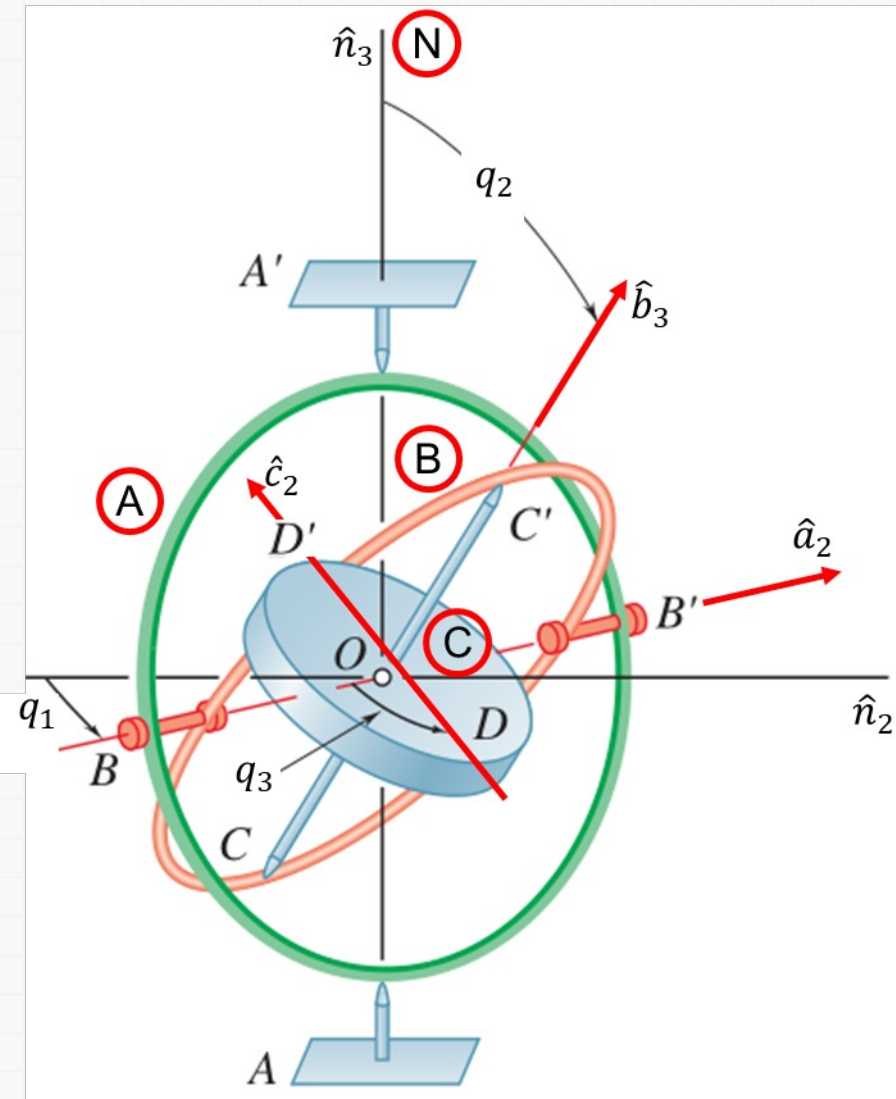
$$\vec{M}_G = \frac{1}{2} m r^2 \dot{q}_1 \dot{q}_3 \hat{b}_2 = I_Z {}^N \vec{\omega}^A \times {}^B \vec{\omega}^C$$

When $q_2 = 90^\circ$, the couple $\vec{M}_G$ is relatively large to change the orientation of their axes.

Therefore, gyroscopes are used as stabilizers in torpedoes and ships.

Spinning bullets and shells remain tangent to their trajectory because of gyroscopic action.

Bicycle is easier to keep balanced at high speeds.



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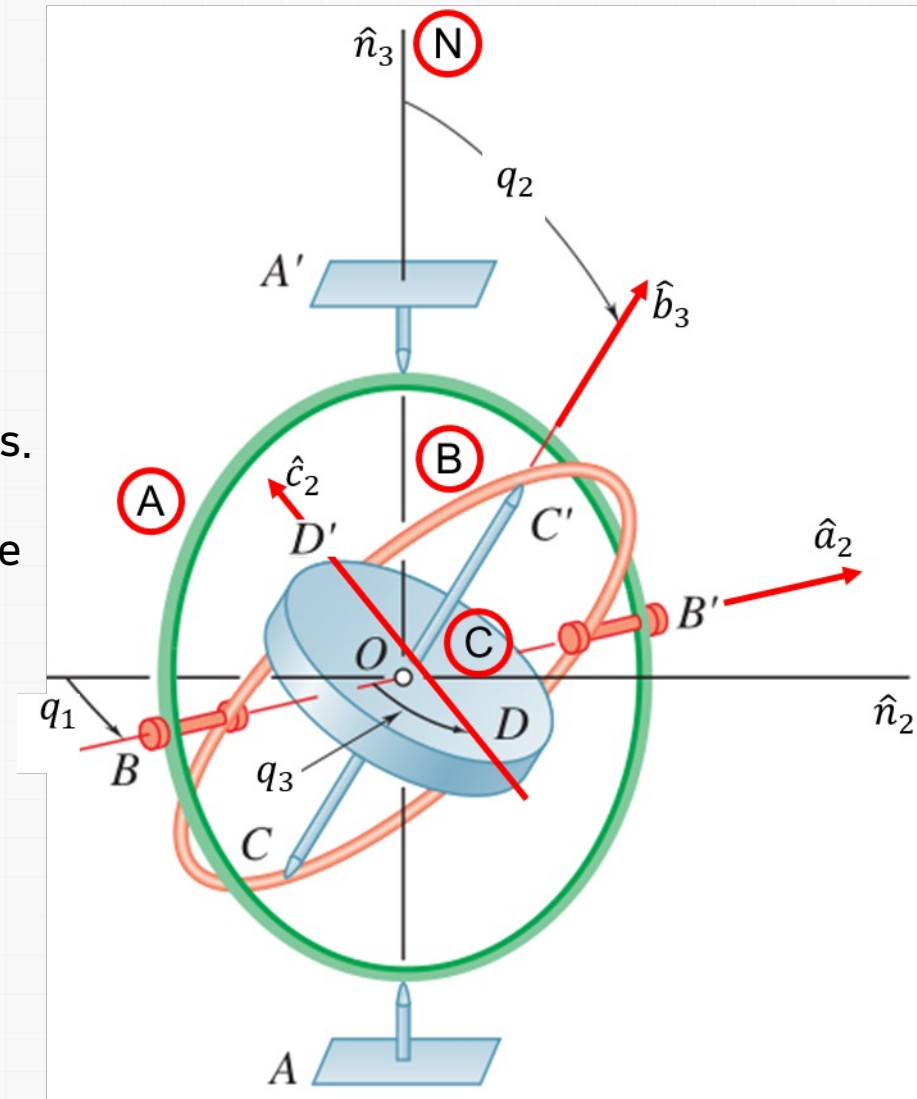
Usage of a Gyroscope

Gyroscopes are used to indicate direction.

- Aircraft uses a magnetic compass to know the heading direction. However, due to inherent characteristics of magnetic fields and compass, heading direction from the compass may not be accurate. Since gyroscope maintains its orientation in space, it can provide directional information to aircraft.

Gyroscope is not affected by the conditions that compromise compass. Therefore, gyroscope can complement the magnetic compass.

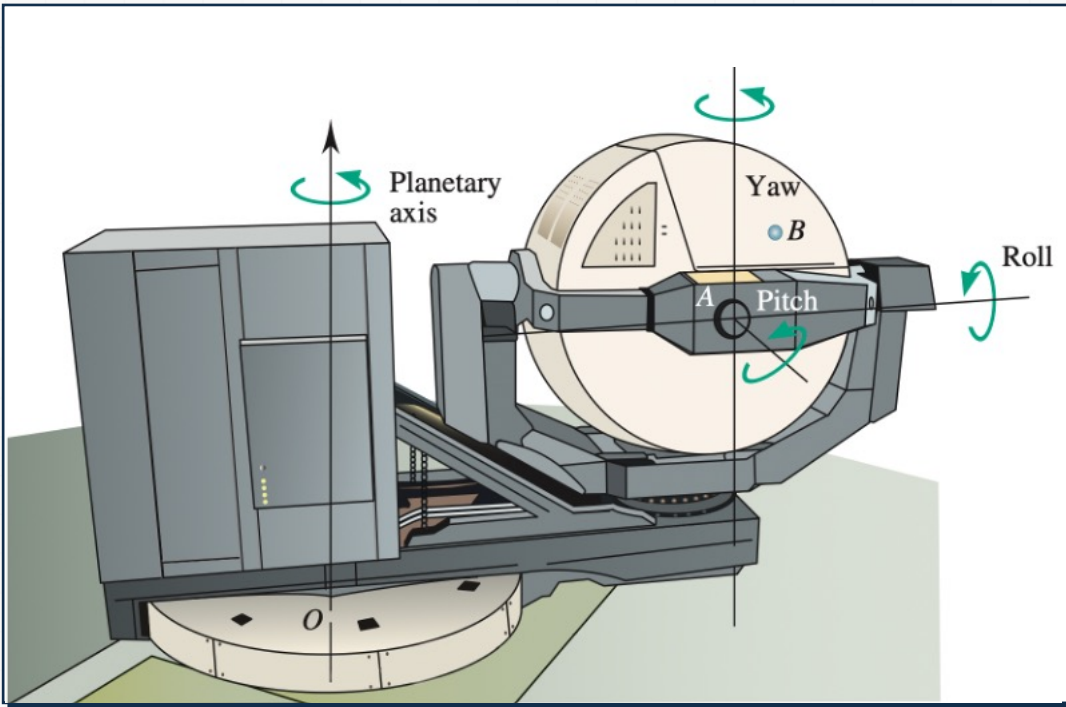
Due to mechanical friction, however, gyroscope precesses. This can be fixed by realigning the direction from gyroscope every 10-15 min.



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Sample Problem 1

⚙️ Consider the following flight simulator.



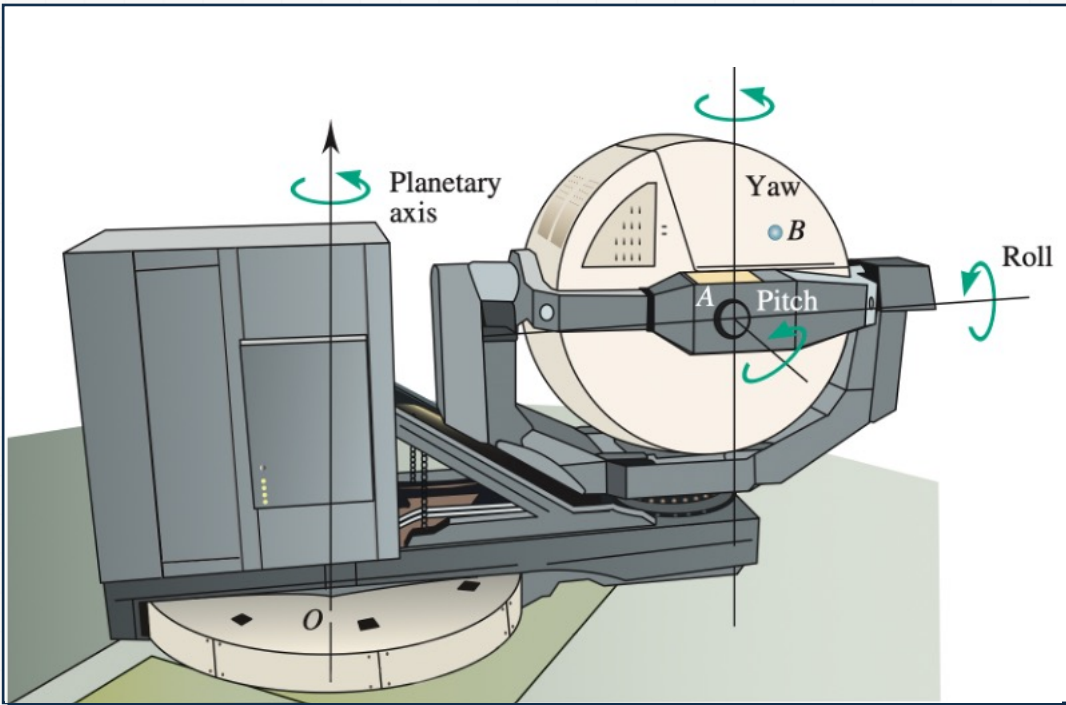
• Consider

- Motion-based flight simulators can help train pilots to recognize situations in which they may become disoriented. The simulator has four degrees of freedom and can rotate a full 400° about the planetary, the roll (x), the yaw (y), and the pitch (z) axes. The principal axes are aligned with the xyz on the figure and $I_x = 400 \text{ kg}\cdot\text{m}^2$, $I_y = 450 \text{ kg}\cdot\text{m}^2$, and $I_z = 900 \text{ kg}\cdot\text{m}^2$.

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Sample Problem 1

⚙️ Consider the following flight simulator.



• Determine

- At the instant shown, the gondola is rotating at a constant 30 rpm about the roll axis, and 20 rpm about the pitch axis. The angular velocity about the pitch axis is increasing at a rate of $90^\circ/s^2$.
- Determine the dynamic loads at the roll gimbals at the instant shown.

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Sample Problem 1

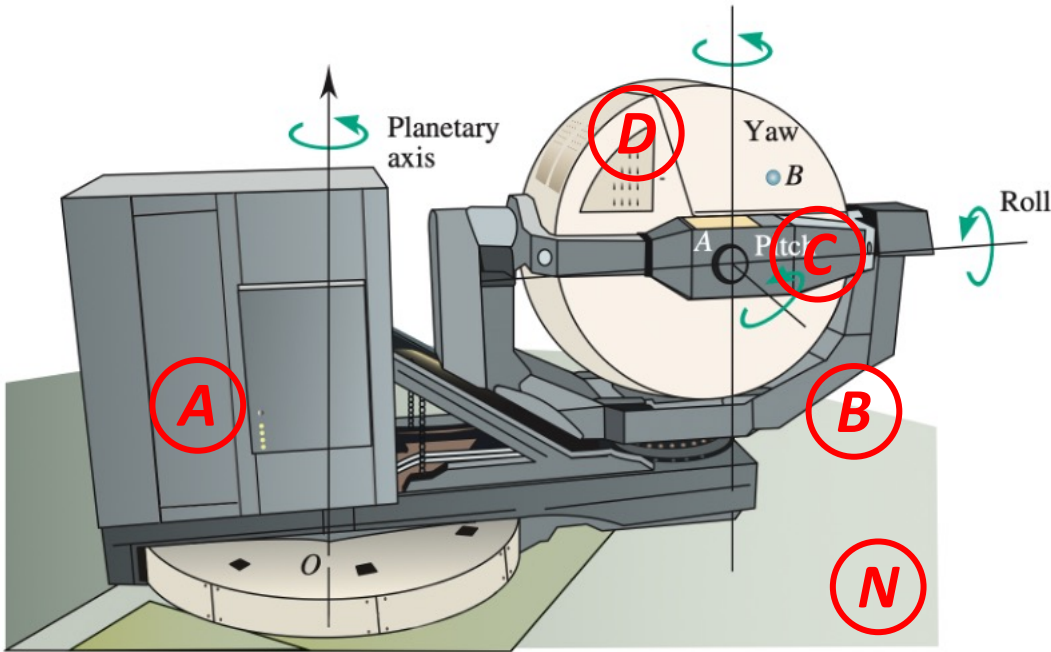
⚙️ **Consider the following flight simulator.**

• **Strategy**

- Identify appropriate reference frames.
- Assign appropriate coordinate systems to each reference frame.
- Assign appropriate generalized coordinates.
- Build the direction cosine table by performing necessary dot products.
- Compose the angular velocities using simple angular velocities.
- Compute the angular momentum.
- Derive the equations of motion.

⚙️ Identify reference frames.

- Each rigid body has one reference frame



Inertial reference frame: N

Reference frame of cube: A

Reference frame for yaw motion: B

Reference frame for roll motion: $\mathcal{C}$

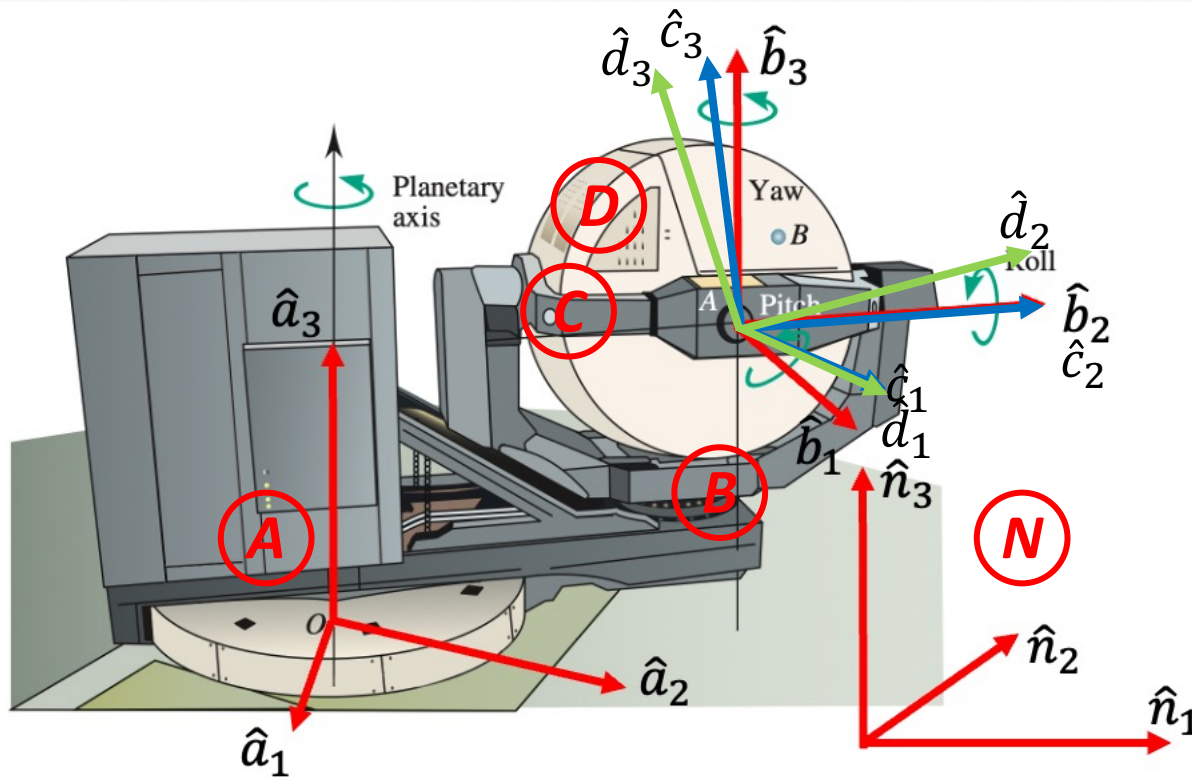
Reference frame for pitch motion: D

16 Gyroscopic Motion

Sample Problem 1

⚙️ **Assign a coordinate system to each reference frame.**

- Each reference frame has at least one coordinate system



Inertial reference frame: N

$\hat{n}_1 \hat{n}_2 \hat{n}_3$

Reference frame of cube: A

$\hat{a}_1 \hat{a}_2 \hat{a}_3$

Reference frame for yaw motion: B

$\hat{b}_1 \hat{b}_2 \hat{b}_3$

Reference frame for roll motion: C

$\hat{c}_1 \hat{c}_2 \hat{c}_3$

Reference frame for pitch motion: D

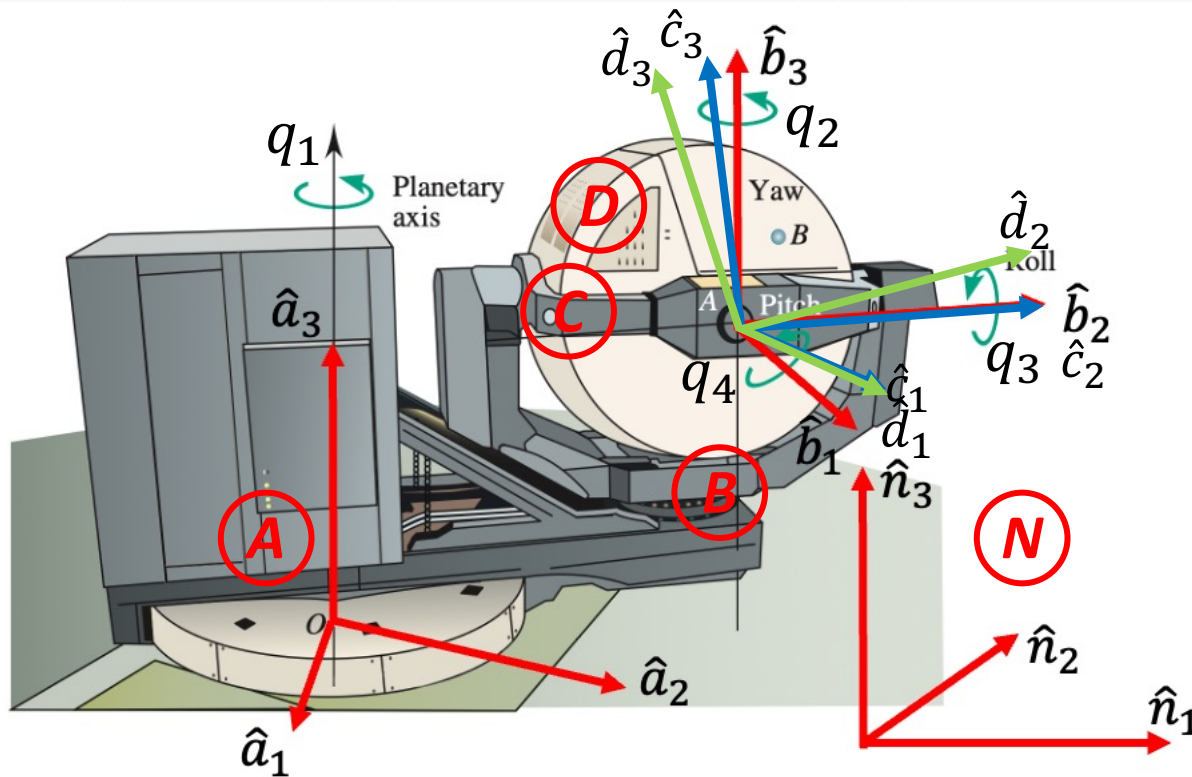
$\hat{d}_1 \hat{d}_2 \hat{d}_3$

16 Gyroscopic Motion

Sample Problem 1

⚙️ Assign generalized coordinates.

- Each single DOF has one generalized coordinate.



$$q_1: \hat{n}_1 \hat{n}_2 \hat{n}_3 \rightarrow \hat{a}_1 \hat{a}_2 \hat{a}_3$$

$$q_2: \hat{a}_1 \hat{a}_2 \hat{a}_3 \rightarrow \hat{b}_1 \hat{b}_2 \hat{b}_3$$

$$q_3: \hat{b}_1 \hat{b}_2 \hat{b}_3 \rightarrow \hat{c}_1 \hat{c}_2 \hat{c}_3$$

$$q_4: \hat{c}_1 \hat{c}_2 \hat{c}_3 \rightarrow \hat{d}_1 \hat{d}_2 \hat{d}_3$$

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Sample Problem 1

⚙️ **Generate direction cosine tables (using rotation matrices)**

- Use rotation matrix for each simple axis of rotation.

	$\hat{a}_1$	$\hat{a}_2$	$\hat{a}_3$
$\hat{n}_1$	$\cos q_1$	$-\sin q_1$	0
$\hat{n}_2$	$\sin q_1$	$\cos q_1$	0
$\hat{n}_3$	0	0	1

Rotation about z axis by q_1

	$\hat{c}_1$	$\hat{c}_2$	$\hat{c}_3$
$\hat{b}_1$	$\cos q_3$	0	$\sin q_3$
$\hat{b}_2$	0	1	0
$\hat{b}_3$	$-\sin q_3$	0	$\cos q_3$

Rotation about y axis by q_3

	$\hat{b}_1$	$\hat{b}_2$	$\hat{b}_3$
$\hat{a}_1$	$\cos q_2$	$-\sin q_2$	0
$\hat{a}_2$	$\sin q_2$	$\cos q_2$	0
$\hat{a}_3$	0	0	1

Rotation about z axis by q_2

	$\hat{d}_1$	$\hat{d}_2$	$\hat{d}_3$
$\hat{c}_1$	1	0	0
$\hat{c}_2$	0	$\cos q_4$	$-\sin q_4$
$\hat{c}_3$	0	$\sin q_4$	$\cos q_4$

Rotation about x axis by q_4

$$q_1: \hat{n}_1 \hat{n}_2 \underline{\hat{n}_3} \rightarrow \hat{a}_1 \hat{a}_2 \underline{\hat{a}_3}$$

$$q_2: \hat{a}_1 \hat{a}_2 \underline{\hat{a}_3} \rightarrow \hat{b}_1 \hat{b}_2 \underline{\hat{b}_3}$$

$$q_3: \hat{b}_1 \underline{\hat{b}_2} \hat{b}_3 \rightarrow \hat{c}_1 \underline{\hat{c}_2} \hat{c}_3$$

$$q_4: \underline{\hat{c}_1} \hat{c}_2 \hat{c}_3 \rightarrow \underline{\hat{d}_1} \hat{d}_2 \hat{d}_3$$

16 Gyroscopic Motion

Sample Problem 1

⚙️ Compute the angular velocities.

- Compose the angular velocities with simple ones and addition theorem.

$${}^N\vec{\omega}^A = \dot{q}_1 \hat{n}_3$$

$${}^A\vec{\omega}^B = \dot{q}_2 \hat{a}_3$$

$${}^B\vec{\omega}^C = \dot{q}_3 \hat{b}_2$$

$${}^C\vec{\omega}^D = \dot{q}_4 \hat{c}_1$$

$$\begin{aligned} {}^N\vec{\omega}^D &= {}^N\vec{\omega}^A + {}^A\vec{\omega}^B + {}^B\vec{\omega}^C + {}^C\vec{\omega}^D \\ &= \dot{q}_1 \hat{n}_3 + \dot{q}_2 \hat{a}_3 + \dot{q}_3 \hat{b}_2 + \dot{q}_4 \hat{c}_1 \end{aligned}$$

$$q_1: \hat{n}_1 \hat{n}_2 \underline{\hat{n}_3} \rightarrow \hat{a}_1 \hat{a}_2 \underline{\hat{a}_3}$$

$$q_2: \hat{a}_1 \hat{a}_2 \underline{\hat{a}_3} \rightarrow \hat{b}_1 \hat{b}_2 \underline{\hat{b}_3}$$

$$q_3: \hat{b}_1 \underline{\hat{b}_2} \hat{b}_3 \rightarrow \hat{c}_1 \underline{\hat{c}_2} \hat{c}_3$$

$$q_4: \underline{\hat{c}_1} \hat{c}_2 \hat{c}_3 \rightarrow \underline{\hat{d}_1} \hat{d}_2 \hat{d}_3$$

Sample Problem 1

⚙️ **Express the angular velocity of gondola in a simpler form.**

- A coordinate system can express a vector in a simpler form the others.

$$\begin{aligned} {}^N\vec{\omega}^D &= \dot{q}_1 \hat{n}_3 + \dot{q}_2 \hat{a}_3 + \dot{q}_3 \hat{b}_2 + \dot{q}_4 \hat{c}_1 \\ &= (-\dot{q}_1 \sin q_3 - \dot{q}_2 \sin q_3 + \dot{q}_4) \hat{d}_1 \\ &\quad + (\dot{q}_1 \cos q_3 \sin q_4 + \dot{q}_2 \cos q_3 \sin q_4 + \dot{q}_3 \cos q_4) \hat{d}_2 \\ &\quad + (\dot{q}_1 \cos q_3 \cos q_4 + \dot{q}_2 \cos q_3 \cos q_4 - \dot{q}_3 \sin q_4) \hat{d}_3 \\ &= (-\dot{q}_1 \sin q_3 - \dot{q}_2 \sin q_3 + \dot{q}_4) \hat{c}_1 \\ &\quad + \dot{q}_3 \hat{c}_2 \\ &\quad + (\dot{q}_1 + \dot{q}_2) \cos q_3 \hat{c}_3 \end{aligned}$$

16 Gyroscopic Motion

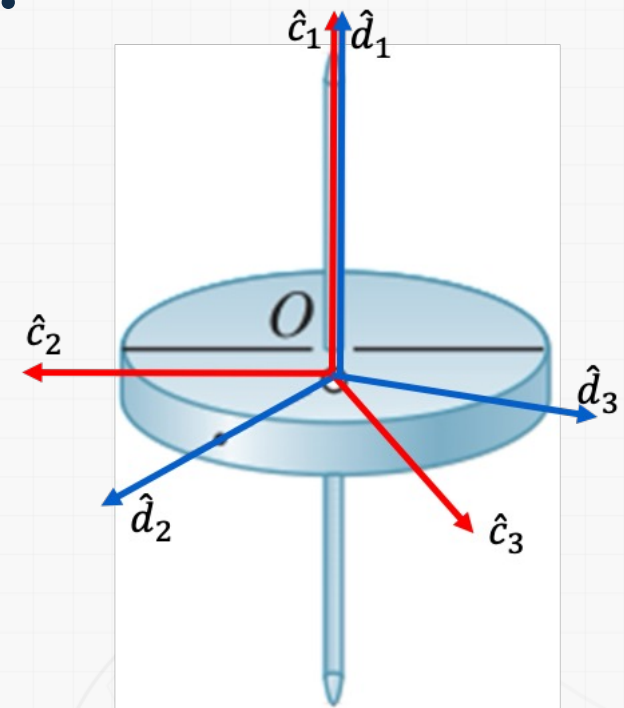
Sample Problem 1

⚙️ Compute the moment of inertia of gondola.

- Moment of inertia of rotor about COM in $\hat{d}_1\hat{d}_2\hat{d}_3$

$$I_G = \begin{bmatrix} 900 & 0 & 0 \\ 0 & 400 & 0 \\ 0 & 0 & 450 \end{bmatrix}$$

Note that $(I_G)_{\hat{d}_1\hat{d}_2\hat{d}_3} \neq (I_G)_{\hat{c}_1\hat{c}_2\hat{c}_3}$



16 Gyroscopic Motion

Sample Problem 1

⚙️ Compute angular momentum of gondola

- Angular momentum (in terms of body reference frame $\hat{d}_1 \hat{d}_2 \hat{d}_3$)

$$\begin{aligned}
 {}^N \vec{H}_G^D &= I_G {}^N \vec{\omega}^D \\
 &= \begin{bmatrix} 900 & 0 & 0 \\ 0 & 400 & 0 \\ 0 & 0 & 450 \end{bmatrix} \begin{bmatrix} -\dot{q}_1 \sin q_3 - \dot{q}_2 \sin q_3 + \dot{q}_4 \\ \dot{q}_1 \cos q_3 \sin q_4 + \dot{q}_2 \cos q_3 \sin q_4 + \dot{q}_3 \cos q_4 \\ \dot{q}_1 \cos q_3 \cos q_4 + \dot{q}_2 \cos q_3 \cos q_4 - \dot{q}_3 \sin q_4 \end{bmatrix} \\
 &= 900(-\dot{q}_1 \sin q_3 - \dot{q}_2 \sin q_3 + \dot{q}_4) \hat{d}_1 \\
 &\quad + 400(\dot{q}_1 \cos q_3 \sin q_4 + \dot{q}_2 \cos q_3 \sin q_4 + \dot{q}_3 \cos q_4) \hat{d}_2 \\
 &\quad + 450(\dot{q}_1 \cos q_3 \cos q_4 + \dot{q}_2 \cos q_3 \cos q_4 - \dot{q}_3 \sin q_4) \hat{d}_3
 \end{aligned}$$

Note the abuse of notation

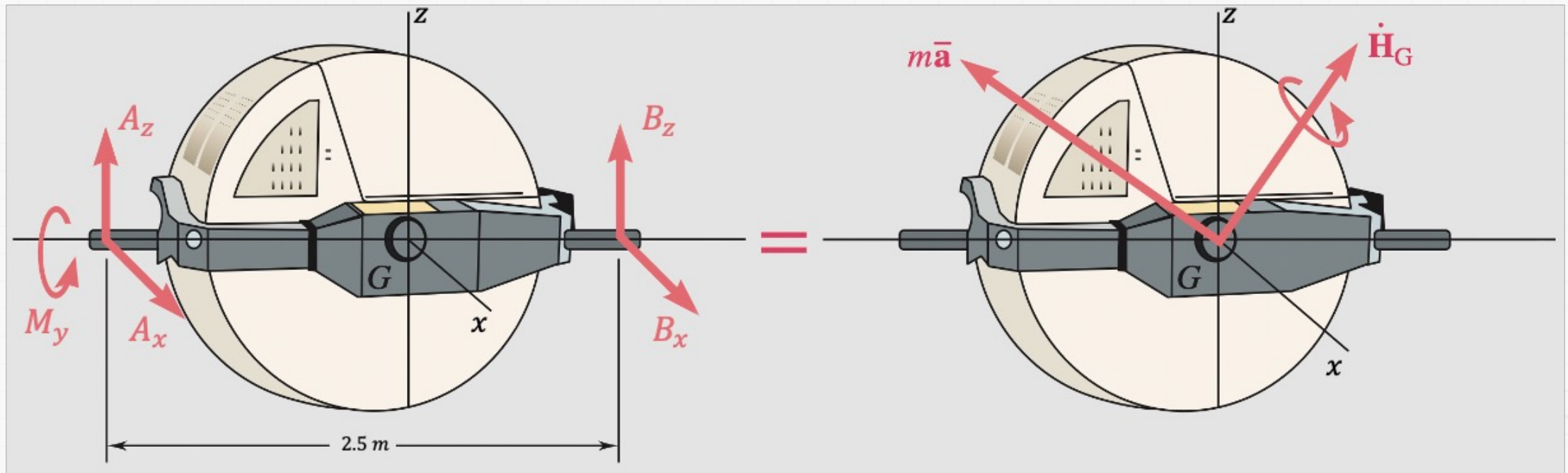
Also note that the above calculation shouldn't be done in terms of $\hat{c}_1 \hat{c}_2 \hat{c}_3$.

16 Gyroscopic Motion

Sample Problem 1

⚙️ Draw free body diagram of gondola and the roll gimbal.

- Assume COM is at the intersection of roll and pitch axes.



16 Gyroscopic Motion

Sample Problem 1

⚙️ Derive equations of motion of gondola.

• Moments

$$\begin{aligned}
\vec{M}_G &= \frac{{}^N d}{{}^N dt} {}^N \vec{H}_G^D = \frac{{}^D d}{{}^D dt} {}^N \vec{H}_G^D + {}^N \vec{\omega}^D \times {}^N \vec{H}_G^D \\
&= 1413.72 \hat{d}_1 - 6250.75 \hat{d}_3 \\
&= M_y \hat{d}_2 - 1.25 \hat{d}_2 \times (A_x \hat{d}_1 + A_z \hat{d}_3) + 1.25 \hat{d}_2 \times (B_x \hat{d}_1 + B_z \hat{d}_3) \\
&= 1.25(B_z - A_z) \hat{d}_1 + M_y \hat{d}_2 + 1.25(A_x - B_x) \hat{d}_3
\end{aligned}$$

$$\dot{q}_3 = 30 \text{rpm} = \pi \frac{\text{rad}}{\text{s}}$$

$$\ddot{q}_3 = 0$$

$$\dot{q}_4 = 20 \text{rpm} = \frac{2\pi \text{ rad}}{3 \text{ s}}$$

$$\ddot{q}_4 = \frac{\pi \text{ rad}}{2 \text{ s}^2}$$

All other variables are zeros

16 Gyroscopic Motion

Sample Problem 1

⚙️ Derive equations of motion of gondola.

• Forces

$$\begin{aligned}\vec{F}_G &= m\vec{a}_G \\ &= 0 \\ &= A_x\hat{d}_1 + A_z\hat{d}_3 + B_x\hat{d}_1 + B_z\hat{d}_3 \\ &= (A_x + B_x)\hat{d}_1 + (A_z + B_z)\hat{d}_3\end{aligned}$$

16 Gyroscopic Motion

Sample Problem 1

⚙️ **Derive equations of motion of gondola.**

- Putting together

$$A_x + B_x = 0$$

$$A_z + B_z = 0$$

$$1.25(B_z - A_z) = 1413.72$$

$$1.25(A_x - B_x) = -6250.75$$

$$M_y = 0$$

↓

$$A_x = -2500.3 \text{ N}$$

$$B_x = 2500.3 \text{ N}$$

$$A_z = -565.49 \text{ N}$$

$$B_z = 565.49 \text{ N}$$

$$M_y = 0 \text{ Nm}$$

16 Gyroscopic Motion

Sample Problem 1

⚙️ Reflect and Think

Dynamic loads on bearings have to be considered when designing equipment with multiple axes of rotation. Products of inertia might introduce additional loads, and could even require additional torque to be input to the motors to obtain desired motion profiles.

16 Gyroscopic Motion

Sample Problem 1

⚙️ Reflect and Think

Also, if you add rotation about the planetary axis, then the linear acceleration of the center of mass will no longer be zero and the angular velocity and angular acceleration will need to be calculated and included in the force equations.

When designing this system, placing the center of mass at the center of rotation and minimizing the products of inertia reduces the reaction forces on the bearings; this is part of the machine balancing process that you typically do in rotating machinery.

References

- Beer, Johnston, Cornwell, Self, Sanghi, "Vector Mechanics for Engineers: Dynamics" 12th Edition, Chapter 18